Pandas Practice Questions — Beginner Level

Answer each question using the dataset below. Try to write the code yourself before checking any solution.

Dataset (use this for all questions)

```
import pandas as pd
import numpy as np

data = {
    'name': ['Amit', 'Riya', 'Kabir', 'Neha', 'Suresh', 'Priya'],
    'age': [25, 30, 22, np.nan, 28, 35],
    'city': ['Delhi', 'Mumbai', 'Pune', 'Delhi', 'Chennai', 'Mumbai'],
    'salary': [40000, 55000, 30000, 45000, 60000, np.nan],
    'department': ['Sales', 'IT', 'Sales', 'HR', 'IT', 'HR']
}
df = pd.DataFrame(data)
```

Q1 — Series

Create a pd.Series directly from the salary values (as a plain list, not from df). Print its dtype, size, and index.

Q2 — DataFrame basics

Print df.info(), df.describe(), df.shape, and df.dtypes.

Q3 — Viewing / Selecting Data

(a) Get the city column using [].
(b) Get the row at position 2 using .iloc[].
(c) Get the row where name is 'Riya' using .loc[].
(d) Get the value of salary for row index 3 using .at[].

Q4 — Basic Operations

(a) Sort df by salary in descending order.
(b) Run df.sort_index() and explain what it does.
(c) Drop the department column (without saving it back).
(d) Rename the column salary to income (without saving it back).
(e) Convert the age column to int using astype() — will this work directly? Why or why not?

Q5 — Filtering Data

(a) Filter rows where city == 'Mumbai'.
(b) Filter rows where city is in ['Delhi', 'Pune'] using .isin().
(c) Filter rows where age is between 25 and 30 using .between().
(d) Filter using .query() for salary > 40000.

Q6 — Handling Missing Data

(a) Check for missing values using isna().
(b) Drop rows with any missing value using dropna().
(c) Fill missing values in age with the column mean using fillna().
(d) Fill missing values in salary using interpolate().

Q7 — GroupBy Basics

(a) Group by department and get the count of employees in each.
(b) Group by department and get the average salary.
(c) Group by department and get min and max age using .agg().

Q8 — Apply & Map

(a) Use .apply() to create a new column bonus = 10% of salary.
(b) Use .map() to convert city values to their first 3 letters.
(c) Write a lambda function inside .apply() to label people as 'Senior' if age > 28, else 'Junior'.

Q9 — Input / Output

(a) Save df to a CSV file called practice.csv without the index.
(b) Read it back into a new DataFrame df2 using read_csv().
(c) Convert df to a dictionary using to_dict().